

Dysprosium-161

¹⁶¹Dy (Dy I), nuclear spin *I* = 5/2. Energy levels and transitions below 2 μm: 39 levels, 55 lines.

Dy + ionisation limit 47901.76 cm⁻¹ = 5.93906 eV (208.761 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element

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J
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e

r

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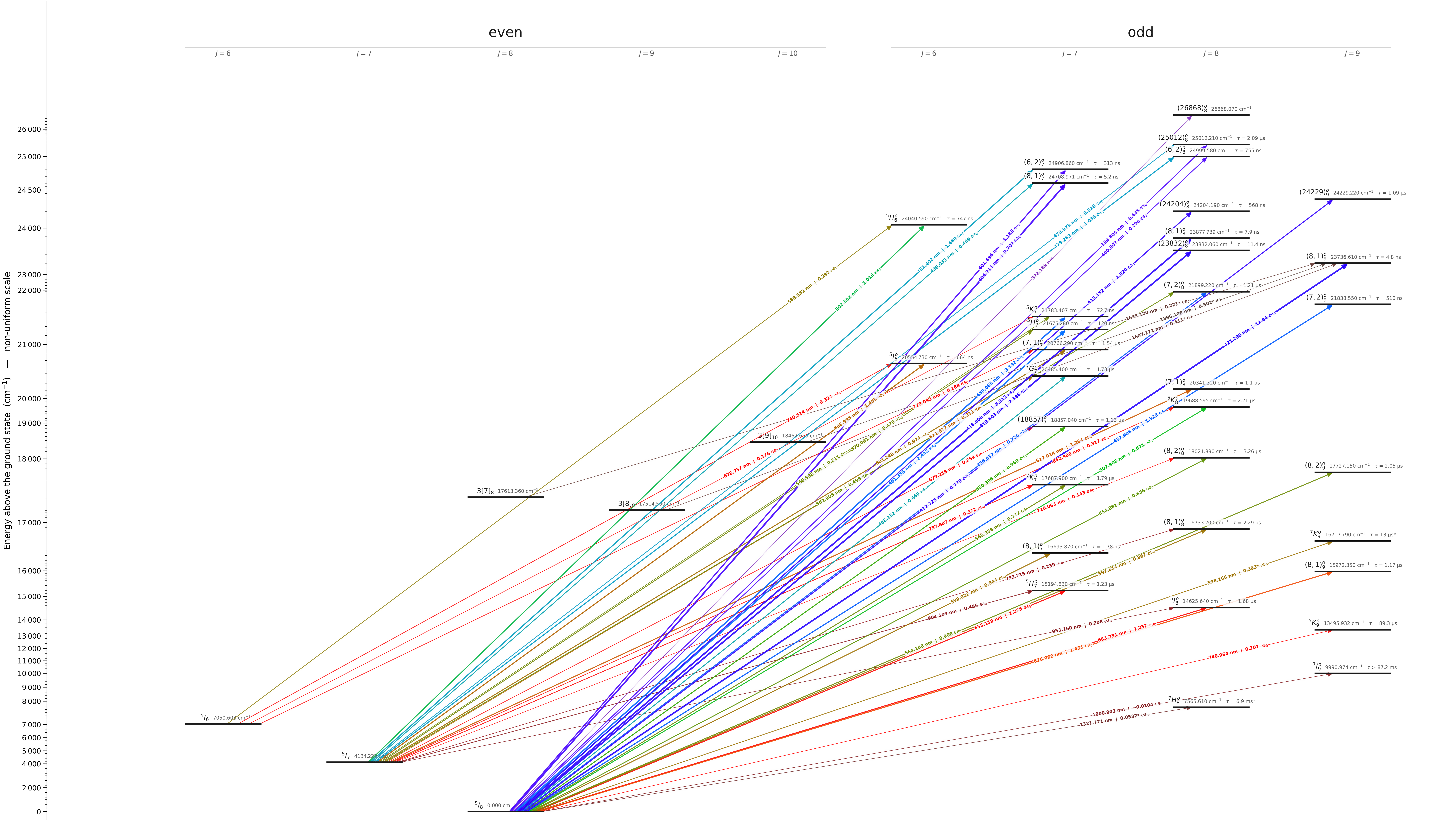
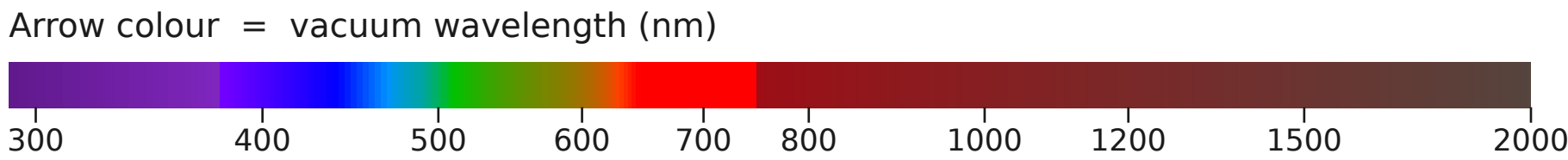
J
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⟩

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{\displaystyle |\langle J|er|J'\rangle |}

 in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[JlerJ'] (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁵ I ₆ – (26868) ⁵ _g	372.1890	372.0832	805.4845	-	-	-	-	UV ground-state line
⁵ I ₆ – (8, 1) ⁵ _g	404.7113	404.5970	740.7563	9.707	1.920e+08	30.61	99.8	strong blue J=8->7 line
⁵ I ₆ – (8, 1) ⁵ _g	418.8001	418.6821	715.8366	8.812	1.260e+08	20.15	99.5	strong blue J=8->8 line
⁵ I ₆ – (23832) ⁵ _g	419.6028	419.4846	714.4672	7.386	8.800e+07	13.96	100	strong blue J=8->8 line
⁵ I ₆ – (8, 1) ⁵ _g	421.2902	421.1715	711.6057	11.84	2.000e+08	33.16	100	main broad cooling line: Zeeman slower, transverse coolin...
⁵ I ₆ – (7, 2) ⁵ _g	456.6373	456.5093	656.5221	0.726	6.600e+05	0.1315	79.9	pure 6s->6p reference line for King plots
⁵ I ₆ – (8, 2) ⁵ _g	564.1065	563.9499	531.4466	0.908	4.900e+05	0.07764	100	closed J=8->9 narrow line
⁵ I ₆ – ⁷ K ^o ₂	598.1652	597.9995	501.1867	0.393*	7.700e+04	0.01224	100	"598 nm" closed J=8->9 narrow line proposed for narrow-li...
⁵ I ₆ – (8, 1) ⁵ _g	599.0223	598.8564	500.4696	0.944	5.600e+05	0.08941	99.7	narrow J=8->7 probe line
⁵ I ₆ – (8, 1) ⁵ _g	626.0819	625.9088	478.8390	1.431	8.900e+05	0.136	100	narrow-line MOT
⁵ I ₆ – ⁵ H ^o ₂	658.1186	657.9368	455.5295	1.275	7.700e+05	0.1294	94.7	narrow probe line used for Doppler thermometry of the 421...
⁵ I ₆ – ⁵ I ^o ₂	683.7308	683.5421	438.4657	1.257	5.893e+05	0.09474	99	narrow J=8->8 optical pumping / demagnetization cooling l...
⁵ I ₆ – ⁵ K ^o ₂	740.9640	740.7600	404.5979	0.207	1.120e+04	0.001782	100	ultra-narrow-line MOT / narrow-line cooling in tweezers
⁵ I ₆ – ⁷ I ^o ₂	1000.9034	1000.6291	299.5219	0.010	1.150e+01	1.825e-06	100	ultra-narrow
⁵ I ₆ – ⁷ H ^o ₂	1321.7705	1321.4091	226.8113	0.053*	1.460e+02	2.307e-05	100	proposed ultra-narrow telecom-band line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
⁵ I ₆ – (26868) ⁵ _g	372.189	164-163	-44(10)
⁵ I ₆ – (26868) ⁵ _g	372.189	164-162	-178(13)
⁵ I ₆ – (26868) ⁵ _g	372.189	164-161	-194(13)
⁵ I ₆ – (26868) ⁵ _g	372.189	164-160	-339(13)
⁵ I ₆ – (8, 1) ⁵ _g	421.290	164-163	-616.3(5)
⁵ I ₆ – (8, 1) ⁵ _g	421.290	164-162	-913.2(8)
⁵ I ₆ – (8, 1) ⁵ _g	421.290	164-161	-1635.0(10)
⁵ I ₆ – (8, 1) ⁵ _g	421.290	164-160	-1895.0(20)
⁵ I ₆ – (8, 1) ⁵ _g	421.290	164-158	-2868(9)
⁵ I ₆ – (8, 1) ⁵ _g	421.290	164-156	-4300(15)
⁵ I ₆ – (7, 2) ⁵ _g	456.637	164-163	-660(3)
⁵ I ₆ – (7, 2) ⁵ _g	456.637	164-162	-971.0(20)
⁵ I ₆ – (7, 2) ⁵ _g	456.637	164-161	-1744(3)
⁵ I ₆ – (7, 2) ⁵ _g	456.637	164-160	-2020(3)
⁵ I ₆ – (7, 2) ⁵ _g	456.637	164-158	-3061(4)
⁵ I ₆ – (7, 2) ⁵ _g	456.637	164-156	-4604(5)
⁵ I ₆ – (8, 2) ⁵ _g	564.106	164-163	-685.0(20)
⁵ I ₆ – (8, 2) ⁵ _g	564.106	163-162	-328(3)
⁵ I ₆ – (8, 2) ⁵ _g	564.106	162-161	-801(3)
⁵ I ₆ – (8, 2) ⁵ _g	564.106	161-160	-284.0(20)
⁵ I ₆ – (8, 2) ⁵ _g	564.106	164-162	-1012.0(20)
⁵ I ₆ – (8, 2) ⁵ _g	564.106	162-160	-1086(5)

Shift = ν(first isotope) – ν(second isotope).

¹⁶¹Dy hyperfine structure (*I* = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
⁵ I ₆	-116.2310(20)	1091.58(5)	F=11/2: +3004.07 F=13/2: +1911.55 F=15/2: +794.21 F=17/2: -286.54 F=19/2: -1261.11 F=21/2: -2051.73
⁵ K ^o ₂	-102.090(20)	3883.0(10)	F=13/2: +3884.65 F=15/2: +1919.82 F=17/2: +210.73 F=19/2: -1048.44 F=21/2: -1640.72 F=23/2: -1326.28
⁵ I ^o ₂	-108.840(20)	2251.0(10)	F=11/2: +3250.82 F=13/2: +1848.36 F=15/2: +525.59 F=17/2: -590.89 F=19/2: -1357.56 F=21/2: -1614.05
(8, 1) ⁵ _g	-187.000	-317.000	F=13/2: +4566.23 F=15/2: +3261.62 F=17/2: +1740.81 F=19/2: -12.07 F=21/2: -2014.73 F=23/2: -4286.75
(8, 1) ⁵ _g	-64.286(17)	892.2(9)	F=9/2: +1619.07 F=11/2: +990.48 F=13/2: +362.32 F=15/2: -208.06 F=17/2: -654.49 F=19/2: -901.96
(8, 2) ⁵ _g	-148.004(13)	1561.0(9)	F=13/2: +4235.74 F=15/2: +2643.63 F=17/2: +1047.38 F=19/2: -474.96 F=21/2: -1836.18 F=23/2: -2939.84
(8, 1) ⁵ _g	-86.900(20)	1747.4(5)	F=13/2: +2772.10 F=15/2: +1580.71 F=17/2: +463.46 F=19/2: -492.29 F=21/2: -1188.89 F=23/2: -1518.40
(26868) ⁵ _g	-78.7(17)	1513(58)	F=11/2: +2309.76 F=13/2: +1331.07 F=15/2: +400.39 F=17/2: -397.16 F=19/2: -965.14 F=21/2: -1195.75

Measured A, B constants; sources in the data file.